FM KERNEL OF THE IDENTITY FUNCTOR IS THE DIAGONAL

THE STACKS PROJECT AUTHORS

Allowed standard prerequisites (editorial).

- Box product surjection. lemma-on-product L26469 (carded this batch). Provides surjections onto the top cohomology sheaf.
- Invertible pushforward detection. functors-lemma-pushforward-invertible-pre (cross-chapter, named). Identifies the kernel as a pushforward from a section.

Source reference. Original reference 0FZP: focused-seeds/geometry/S9-eqv-uniqueness/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0FZ0: sources/stacks-equiv/pinned-original-source.tex, source lines 1750–1750. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let k be a field. Let X be a smooth proper scheme over k. Let $K \in D_{perf}(\mathcal{O}_{X\times X})$. If the Fourier-Mukai functor $\Phi_K : D_{perf}(\mathcal{O}_X) \to D_{perf}(\mathcal{O}_X)$ is isomorphic to the identity functor, then $K \cong \Delta_*\mathcal{O}_X$ in ${}_{perf}(\mathcal{O}_{X\times X})$.*

Proof. Let i be the minimal integer such that the cohomology sheaf $H^i(K)$ is nonzero. Let $\mathcal{E}$ and $\mathcal{G}$ be finite locally free $\mathcal{O}_X$-modules. Then

$$\begin{aligned} H^i(X \times X, K \otimes^{\mathbf{L}}_{\mathcal{O}_{X\times X}} (\mathcal{E} \boxtimes \mathcal{G})) &= H^i(X, R\mathrm{pr}_{2,*}(K \otimes^{\mathbf{L}}_{\mathcal{O}_{X\times X}} (\mathcal{E} \boxtimes \mathcal{G}))) \\ &= H^i(X, \Phi_K(\mathcal{E}) \otimes^{\mathbf{L}}_{\mathcal{O}_X} \mathcal{G}) \\ &\cong H^i(X, \mathcal{E} \otimes \mathcal{G}) \end{aligned}$$

which is zero if $i < 0$. On the other hand, we can choose $\mathcal{E}$ and $\mathcal{G}$ such that there is a surjection $\mathcal{E}^\vee \boxtimes \mathcal{G}^\vee \to H^i(K)$ by Lemma 0FZ0. In this case the left hand side of the equalities is nonzero. Hence we conclude that $H^i(K) = 0$ for $i < 0$.

Let i be the maximal integer such that $H^i(K)$ is nonzero. The same argument with $\mathcal{E}$ and $\mathcal{G}$ support of dimension 0 shows that $i \leq 0$. Hence we conclude that K is given by a single coherent $\mathcal{O}_{X\times X}$-module $\mathcal{K}$ sitting in degree 0.

Since $R\mathrm{pr}_{2,*}(\mathrm{pr}_1^*\mathcal{F} \otimes \mathcal{K})$ is $\mathcal{F}$, by taking $\mathcal{F}$ supported at closed points we see that the support of $\mathcal{K}$ is finite over X via pr_2. Since $R\mathrm{pr}_{2,*}(\mathcal{K}) \cong \mathcal{O}_X$ we conclude by Functors and Morphisms, Lemma 0FZP that $\mathcal{K} = s_*\mathcal{O}_X$ for some section $s : X \to$

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$X \times X$ of the second projection. Then $\Phi_K(M) = f^*M$ where $f = \mathrm{pr}_1 \circ s$ and this can happen only if s is the diagonal morphism as desired. $\square$